

# Moses C. Nah

ROBOTICS, CONTROL THEORY, AND MOTOR NEUROSCIENCE

✉ moses890@gmail.com | 🏠 mosesandlily.github.io | 📺 MosesAndLily

*Mount Everest stands to be the highest since its with the Himalayas.*

## Education

### Massachusetts Institute of Technology (MIT)

PH.D. IN MECHANICAL ENGINEERING

Cambridge, MA, USA

May 2020 – Dec. 2024

- Advisor: Prof. Neville Hogan
- Thesis: [Modular Robot Control with Motor Primitives](#) (Defense: Oct. 29, 2024)
- Committee: Prof. Jean-Jacques Slotine, Prof. Alberto Rodriguez
- Major: System Dynamics and Control
- Minor: Information Theory and Machine Learning

### Massachusetts Institute of Technology (MIT)

M.S. IN MECHANICAL ENGINEERING

Cambridge, MA, USA

Sept. 2018 – May 2020

- Advisor: Prof. Neville Hogan
- Thesis: [Dynamic Primitives Facilitate Manipulating a Whip](#)
- Qualifying Exam Subjects: Dynamics, Control, Stochastic Systems

### Seoul National University (SNU)

B.S. IN MECHANICAL AND AEROSPACE ENGINEERING

Seoul, South Korea

Mar. 2011 – Sept. 2018

- Summa Cum Laude
- Leave of Absence for Military Service (2 years)

### Gyeonggibuk Science High School

EARLY GRADUATION WITH HONORS

Gyeonggi-do, South Korea

Mar. 2009 – Mar. 2011

- Gold Medal, 12th Korean Physics Olympiad (KPhO) [HA13]
- Qualifier, International Physics Olympiad (IPhO) Winter and Summer Schools
- First Place in the Placement Examination [HA14]

## Professional & Research Experience

### Honda Research Institute USA

POSTDOCTORAL RESEARCH SCIENTIST: ROBOTICS AND CONTROL SYSTEMS

San Jose, CA, USA

Aug. 2025 – Present

- Division: Intelligent Robotics Research (IRR)
- Developing intelligent, self-improving control algorithms for contact-rich manipulation tasks

### Massachusetts Institute of Technology (MIT)

POSTDOCTORAL ASSOCIATE

Cambridge, MA, USA

Jan. 2025 – July 2025

- Supervisor: Prof. Neville Hogan
- Teaching Assistant: 2.152 Nonlinear Control (Prof. Jean-Jacques Slotine) [T01]
- UROP Research Advisor. Project: Human Motor Control of Spatial Orientation

### SNU Biorobotics Lab

UNDERGRADUATE RESEARCHER (BACHELOR'S THESIS)

Seoul, S.Korea

Mar. 2017 – Feb. 2018

- Supervisor: Prof. Kyu-Jin Cho
- Project 1: Stabilization of the Underactuation Mechanism of the SNU Exo-Glove Poly
- Project 2: Development of Circuitry for Material Stiffness Identification Using Force-Sensitive Resistors (FSR)

### NAVER LABS Robotics Team

ROBOTICS RESEARCH INTERN

Gyeonggi-do, S.Korea

Aug. 2016 – Feb. 2017

- Supervisor: Dr. Sang-ok Seok
- Project 1: Development of a Wheel-Based Stair-Climbing Robot [C07]
- Project 2: Development of a Li-Ion Battery Pack PCB for Universal Use in NAVER LABS Robots [C06]

# Publications

---

## JOURNAL

- [J01] Johannes Lachner, **Moses C. Nah**, Neville Hogan. *A Physically Consistent Stiffness Formulation for Contact-Rich Manipulation*. The International Journal of Robotics Research (IJRR). 2025
- [J02] **Moses C. Nah**, Johannes Lachner, Neville Hogan. *Robot Control Based on Motor Primitives: A Comparison of Two Approaches*. The International Journal of Robotics Research (IJRR). 2024
- [J03] **Moses C. Nah**, Aleksei Krotov, Marta Russo, Dagmar Sternad, Neville Hogan. *Learning to Manipulate a Whip with Simple Primitive Actions: A Simulation Study*. iScience. 2023
- [J04] Aleksei Krotov, Marta Russo, **Moses C. Nah**, Neville Hogan, Dagmar Sternad. *Motor Control Beyond Reach: How Humans Hit a Target with a Whip*. Royal Society Open Science. 2022

## CONFERENCE

- [C01] **Moses C. Nah**, Johannes Lachner, Federico Tessari, Neville Hogan. *On the Modularity of Elementary Dynamic Actions*. International Conference on Intelligent Robots and Systems (IROS). 2024. **Best Conference Paper Award [HA01]**
- [C02] Johannes Lachner\*, **Moses C. Nah\***, Stefano Stramigioli, Neville Hogan. *Exp[licit]: An Educational Robot Modeling Software Based on Exponential Maps*. International Conference on Advanced Intelligent Mechatronics (AIM). 2024. **\*Equal Contribution**. <https://github.com/explicit-robotics>
- [C03] **Moses C. Nah**, Aleksei Krotov, Marta Russo, Dagmar Sternad, Neville Hogan. *Manipulating a Whip in 3D via Dynamic Primitives*. International Conference on Intelligent Robots and Systems (IROS). 2021
- [C04] Xiaofeng Xiong, **Moses C. Nah**, Aleksei Krotov, Dagmar Sternad. *Online Impedance Adaptation Facilitates Manipulating a Whip*. International Conference on Intelligent Robots and Systems (IROS). 2021
- [C05] **Moses C. Nah**, Aleksei Krotov, Marta Russo, Dagmar Sternad, Neville Hogan. *Dynamic Primitives Facilitate Manipulating a Whip*. International Conference on Biomedical Robotics and Biomechatronics (BIOROB). 2020. **Best Student Paper Award [HA04]**
- [C06] Dongil Choi, Minsu Kim, Hyeongkeun Kim, Choe Jonghun, **Moses C. Nah**. *Motion Planning of Autonomous Personal Transporter Using Model Predictive Control for Minimizing Non-Minimum Phase Behavior*. International Conference on Ubiquitous Robots (UR). 2018
- [C07] Jonghun Choe\*, Ukjin Kwon\*, **Moses C. Nah\***, Hyeongkeun Kim\*. *Design Analysis of Tuskbot: Universal Stair-Climbing 4-Wheel Indoor Robot*. International Conference on Intelligent Robots and Systems (IROS). 2017. **\*Equal Contribution**.

## ARXIV

- [ArX01] **Moses C. Nah**, Johannes Lachner, Neville Hogan. *Modular Robot Control with Motor Primitives*. 2025
- [ArX02] **Moses C. Nah**, Johannes Lachner, Neville Hogan, Jean-Jacques Slotine. *Combining Movement Primitives with Contraction Theory*. 2025
- [ArX03] Johannes Lachner, Federico Tessari, A. Michael West Jr., **Moses C. Nah**, Neville Hogan. *Divide et Impera: Decoding Impedance Strategies for Robotic Peg-in-Hole Assembly*. 2024

# Honors & Awards

---

- 2024 [HA01] **Best Conference Paper Award**, IEEE/RSJ IROS (\$2,000)
- 2024 [HA02] **Sontheimer Travel Award**, Massachusetts Institute of Technology (\$1,500)
- 2023 [HA03] **MathWorks Fellowship**, Massachusetts Institute of Technology
- 2020 [HA04] **Best Student Paper Award**, IEEE BIOROB
- 2020 [HA05] **Presentation Award (First Place)**, MIT MERE (\$500)
- 2018 [HA06] **Startup Investment Award**, SNU Startup Camp (₩1,500,000)
- 2017 [HA07] **Excellence Award**, SNU Smart Social Contribution Contest (₩1,500,000)
- 2017 [HA08] **Excellent Paper Award**, SNU CTL 26<sup>th</sup> Best Report Contest (₩500,000)
- 2017 [HA09] **Young Talent Support Fellowship**, NAVER LABS
- 2017 [HA10] **Gwanak Fellowship**, Hanil Corporation
- 2013 [HA11] **Grand Prize**, SNU Design, Manufacturing Process, and Laboratory Contest
- 2011 [HA12] **First Runner-Up**, SNU Creative Engineering and Design Contest
- 2009 [HA13] **Gold Medalist**, 12<sup>th</sup> Korea Physics Olympiad (KPhO), High School Division
- 2009 [HA14] **First Place (Top Admitted Student)**, Gyeonggibuk Science High School Placement Examination
- 2008 [HA15] **Silver Medalist**, 11<sup>th</sup> Korea Physics Olympiad (KPhO), Middle School Division

## Teaching Experience

---

### Teaching Assistant (MIT)

- Spring 2025 [T01] **2.152 Nonlinear Control**, Prof. Jean-Jacques Slotine (Rating: 7.0/7.0, Median)  
Fall 2024 [T02] **2.151 Advanced System Dynamics and Control**, Prof. Neville Hogan (Rating: 6.9/7.0)  
Fall 2022 [T03] **2.032 Dynamics**, Prof. Triantaphyllos Akylas (Rating: 6.8/7.0)  
Fall 2021 [T04] **2.151 Advanced System Dynamics and Control**, Prof. Neville Hogan (Rating: 6.8/7.0)

### Undergraduate Research Mentor (MIT)

- Jan. 2025–Present [T05] **Bronwyn Busby**, Human Control of Spatial Orientation  
Sept. 2022–June 2023 [T06] **Yinkai Dong**, Impedance Parameter Estimation via Adaptive Sliding Mode Control

## Presentations and Invited Talks

---

### Conference Presentations

- 2023 [P01] **IROS**, Workshop on Leveraging Models for Contact-Rich Manipulation (*OpenReview*)  
2020 [P02] **Neural Control of Movement (NCM)**, Blitz Talk (*Program*)  
2020 [P03] **Neuromatch Conference 3.0**, *Watch Video*

### Poster Presentations

- 2025 [P04] **Harvard NSF Workshop on Reinforcement Learning**  
2021 [P05] **Robotics: Science and Systems (RSS)**, Workshop on Deformable Object Simulation (DO-Sim)  
2020 [P06] **IROS**, Workshop on Impedance Learning (*Website / Abstract*)  
2019 [P07] **MIT Embodied Intelligence Research Mixer**

### Invited Talks

- 2025 [P08] **North Carolina State University**, Topic: Modular Robot Control with Motor Primitives  
2022 [P09] **KUKA Robotics**, Topic: Robot Control Based on Motor Primitives (Virtual)  
2017 [P10] **TEDxSNU**, Featured Speaker, Session: *People Who Make Something*

## Academic Service

---

- Reviewer: Conferences** IROS (2021, 2024, 2025), ICRA (2022, 2024, 2025), BIOROB (2024), ISRR (2024)  
**Reviewer: Journals** T-RO (2024), RA-L (2024)

## Skills

---

- |                                      |                                                                                      |
|--------------------------------------|--------------------------------------------------------------------------------------|
| <b>Robotics &amp; Mechatronics</b>   | KUKA LBR iiwa, Franka Emika Panda, Allegro Hand, ROS 1/2, LabVIEW, KiCAD, SolidWorks |
| <b>Simulation &amp; Modeling</b>     | MuJoCo, MATLAB                                                                       |
| <b>Programming &amp; Tools</b>       | C/C++, Java, Python, Docker, Bash, Git, $\LaTeX$                                     |
| <b>Mathematics (Selected Topics)</b> | Nonlinear Control Theory, Differential Geometry, Rigid Body Kinematics and Dynamics  |
| <b>Languages</b>                     | Korean (Native), English (Full Professional Proficiency), Japanese (Elementary)      |